



Developpement JAR

Développement

```
▼ jar
  towerdefense-a-interactive-music.jar
  towerdefense-a-interactive.jar
  towerdefense-a-random.jar
  towerdefense-b-interactive-music.jar
  towerdefense-b-interactive.jar
  towerdefense-b-random.jar
```

TESTS

Tests

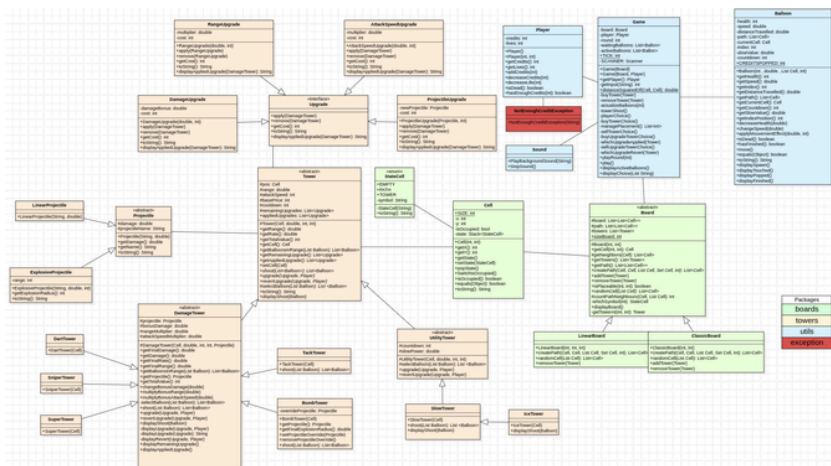
```
JUnit Jupiter ✓
├── AttackSpeedUpgradeTest ✓
│   ├── applyTest() ✓
│   └── removeTest() ✓
├── TackTowerTest ✓
│   └── shootTest() ✓
├── UtilityTowerTest ✓
│   ├── shootTest() ✓
│   ├── upgradeExceptionTest() ✓
│   ├── revertUpgradeExceptionTest() ✓
│   └── selectBalloonTest() ✓
├── GameTest ✓
│   ├── DistanceSquaredOffTest() ✓
│   ├── buyTowerTest() ✓
│   ├── IceTowerTest ✓
│   │   └── applyFreezeTest() ✓
│   ├── SlowTowerTest ✓
│   │   ├── shootTest() ✓
│   │   ├── applySlowTest() ✓
│   │   └── cooldownTest() ✓
│   └── TowerTest ✓
│       ├── upgradeTest() ✓
│       ├── upgradeNoCreditsExceptionTest() ✓
│       ├── revertUpgradeNotExistExceptionTest() ✓
│       ├── upgradeNotExistExceptionTest() ✓
│       ├── revertUpgradeTest() ✓
│       ├── setCellTest() ✓
│       ├── getBalloonsInRangeTest() ✓
│       └── getBalloonsOutOfRangeTest() ✓
├── CellTest ✓
│   ├── testAddInStackState() ✓
│   ├── testIfOccupiedSwitched() ✓
│   └── testIfTurnCellIsAlreadyFull() ✓
└──
```

```
FlappyTest ✓
├── hasEnoughCreditsTest() ✓
├── isDeadTest() ✓
├── decreaseLifeTest() ✓
├── decreaseCreditsTest() ✓
├── addCreditsTest() ✓
├── LinearBoardTest ✓
│   ├── createBoardTest() ✓
│   ├── testRemoveTower() ✓
│   ├── isPlacedInTest() ✓
│   ├── randomCellTest() ✓
│   ├── applyTest() ✓
│   ├── removeTest() ✓
│   ├── testRemoveTower() ✓
│   ├── isPlacedInTest() ✓
│   ├── testAddTower() ✓
│   └── nullTowerTest() ✓
├── addDamageTest() ✓
├── cooldownTest() ✓
├── ProjectileUpgradeTest ✓
│   ├── applyTest() ✓
│   └── removeTest() ✓
├── BoardTest ✓
│   └── testIfNoSpawnsWorking() ✓
├── BalloonTest ✓
│   ├── changeSpeedTest() ✓
│   ├── applyMovementEffectTest() ✓
│   ├── isDeadTest() ✓
│   ├── moveTest() ✓
│   ├── hasFinishedTest() ✓
│   ├── decreaseRemainingTest() ✓
│   ├── applyTest() ✓
│   └── removeTest() ✓
├── DamageTowerTest ✓
│   ├── shootTest() ✓
│   ├── upgradeTest() ✓
│   ├── upgradeNoCreditsExceptionTest() ✓
│   ├── getInitialDamageTest() ✓
│   ├── getInitialRangeTest() ✓
│   ├── revertUpgradeNotExistExceptionTest() ✓
│   ├── changeDamageTest() ✓
│   ├── selectBalloonTest() ✓
│   ├── upgradeNotExistExceptionTest() ✓
│   ├── revertUpgradeTest() ✓
│   ├── getInitialDataTest() ✓
│   └── multiShotAttackSpeedTest() ✓
└── JUnit Vintage ✓
    └── JUnit Platform Suite ✓
```

```
Test run finished after 119 ms
[      21 containers found      ]
[      0 containers skipped    ]
[      21 containers started    ]
[      0 containers aborted    ]
[      21 containers successful ]
[      0 containers failed     ]
[      64 tests found          ]
[      0 tests skipped         ]
[      64 tests started        ]
[      0 tests aborted         ]
[      64 tests successful     ]
[      0 tests failed          ]
```

PACKAGES & UML

Packages



Packages

Package

btd.boards

btd.exceptions

btd.towers.projectiles

[btd.towers.towers](#)

[btd.towers.upgrades](#)

btd.utils

FONCTIONNEMENT DE L'EQUIPE

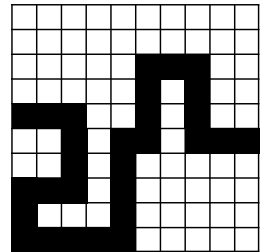
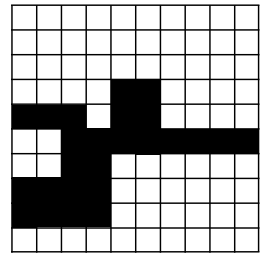
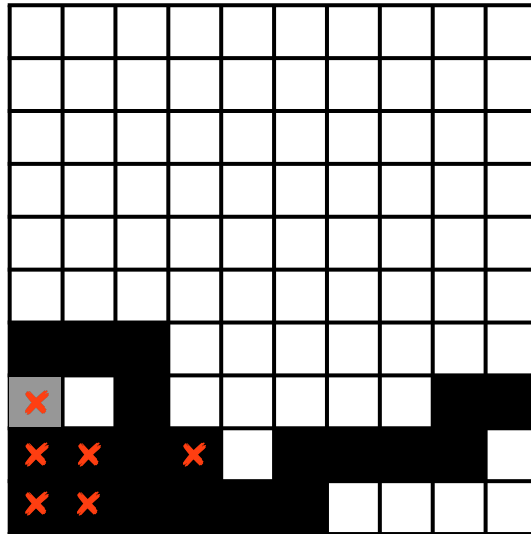
Fonctionnement de l'équipe

Luc	Sami	Kamal	Clément
Chemins	Boards	Balloons	Towers
Gestion tirs	Upgrades	Tests	Game
Interactions joueur	Tests	Musique	Upgrades

LUC
CREATE PATH CLASSIC

Create Path

Classic

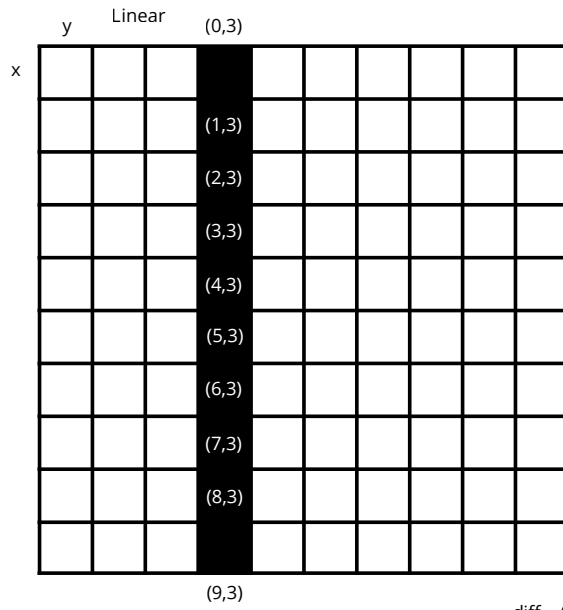


```
public List<Cell> createPath(Cell curr, Cell finale, List<Cell> list);
```

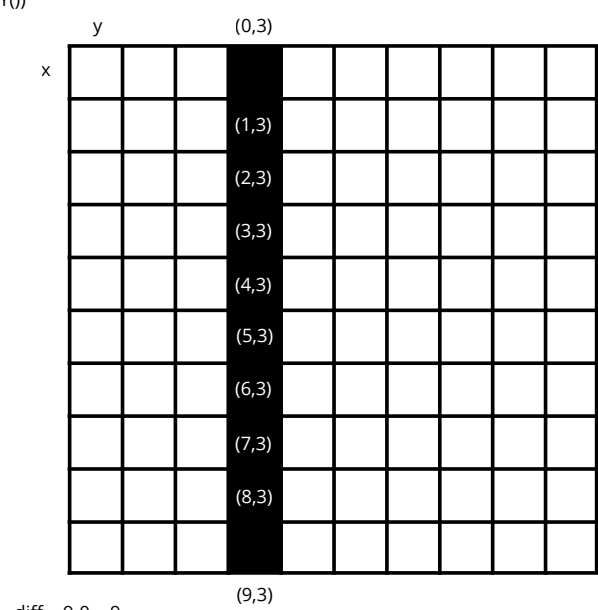
CREATE PATH
LINEAR

Create Path

`curr.getX() + (i * step),
curr.getY())`



$\text{diff} = 0 - 9 = -9$



$\text{diff} = 9 - 0 = 9$

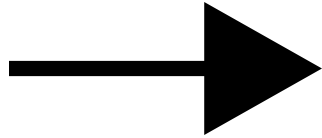
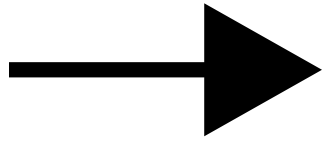
CLEMENT
EXTENSION TOWER

Extensions



EXTENSION UPGRADES

Extensions



POLYVALENCE

Polyvalence

DIFFICULTE GIT

Difficultés principales Gestion Git

1. Les Conflits de Fusion (Merge Conflicts)
2. L'oubli du "Pull" avant le "Push"
3. Travailler tous sur la branche principale

Difficultés principales

Difficultés principales

Calcul du chemin



Refonte graphique du plateau

x \ y	0	1	2	3	4	5	6	7	8	9
0	┌───┐	┌───┐	┌───┐	┌───┐	┌───┐	┌───┐	┌───┐	┌───┐	┌───┐	┌───┐
1				└──┘	└──┘	└──┘				
2										
3	└──┘	└──┘	└──┘							
4				└──┘	└──┘					
5									└──┘	└──┘
6	└──┘	└──┘	└──┘	└──┘				└──┘	└──┘	
7	START									END
8								└──┘	└──┘	└──┘
9										

Gestion des tours / upgrades



BILAN

Bilan



Image générée par IA

NE PAS SLIDE (LIEN DIAPO)

<https://canva.link/i1f0scoufm7g9xd>